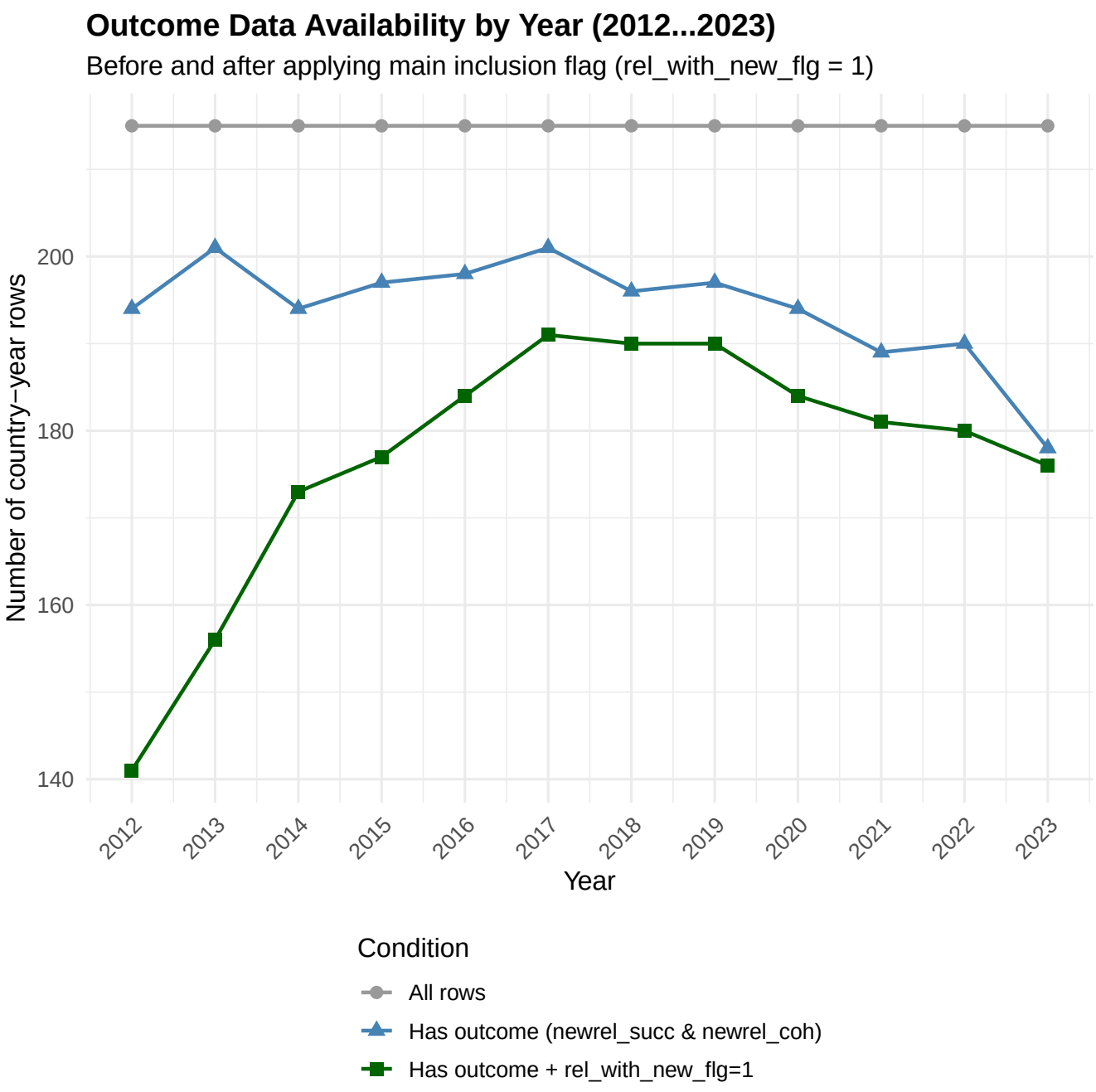
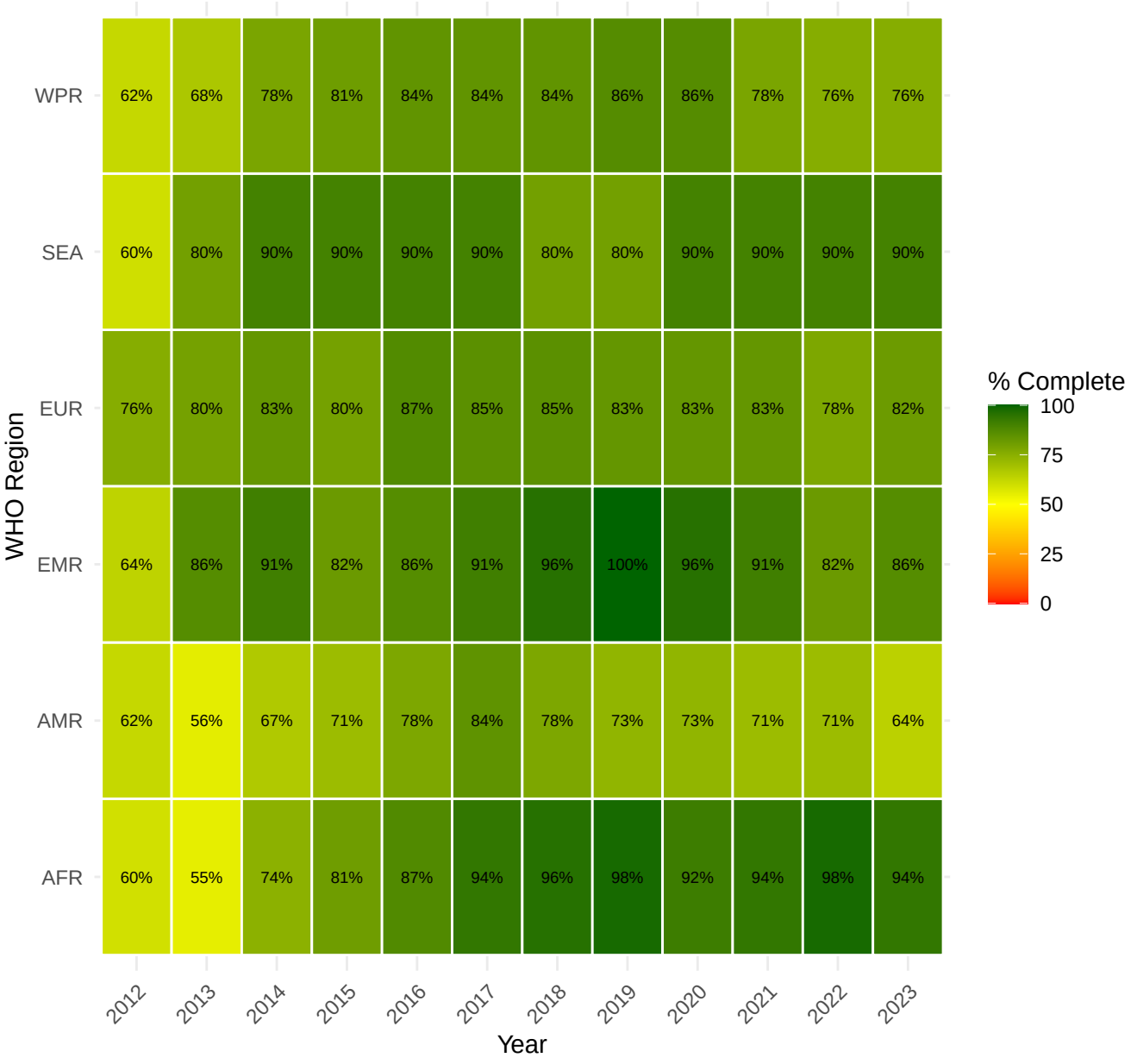
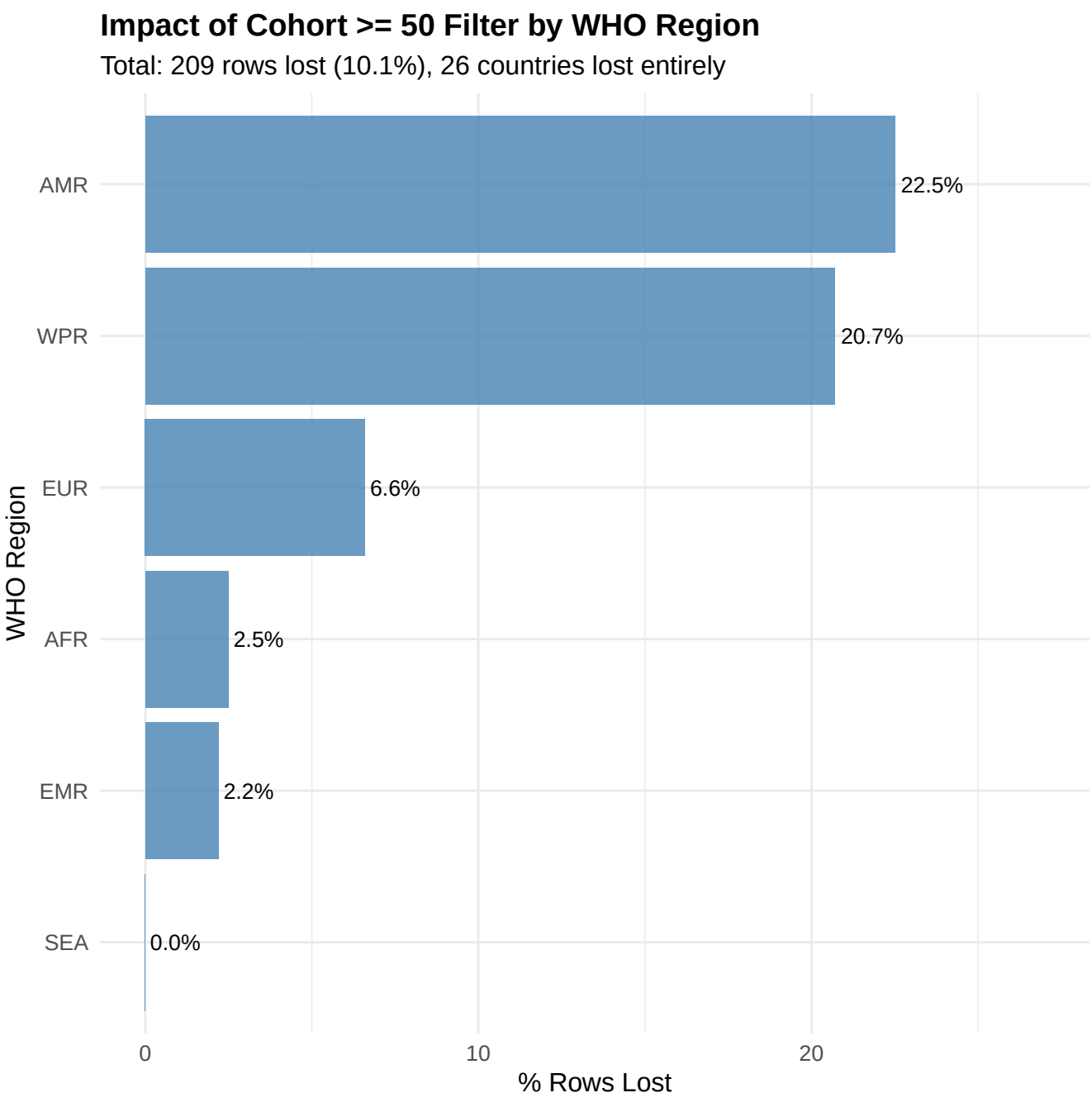
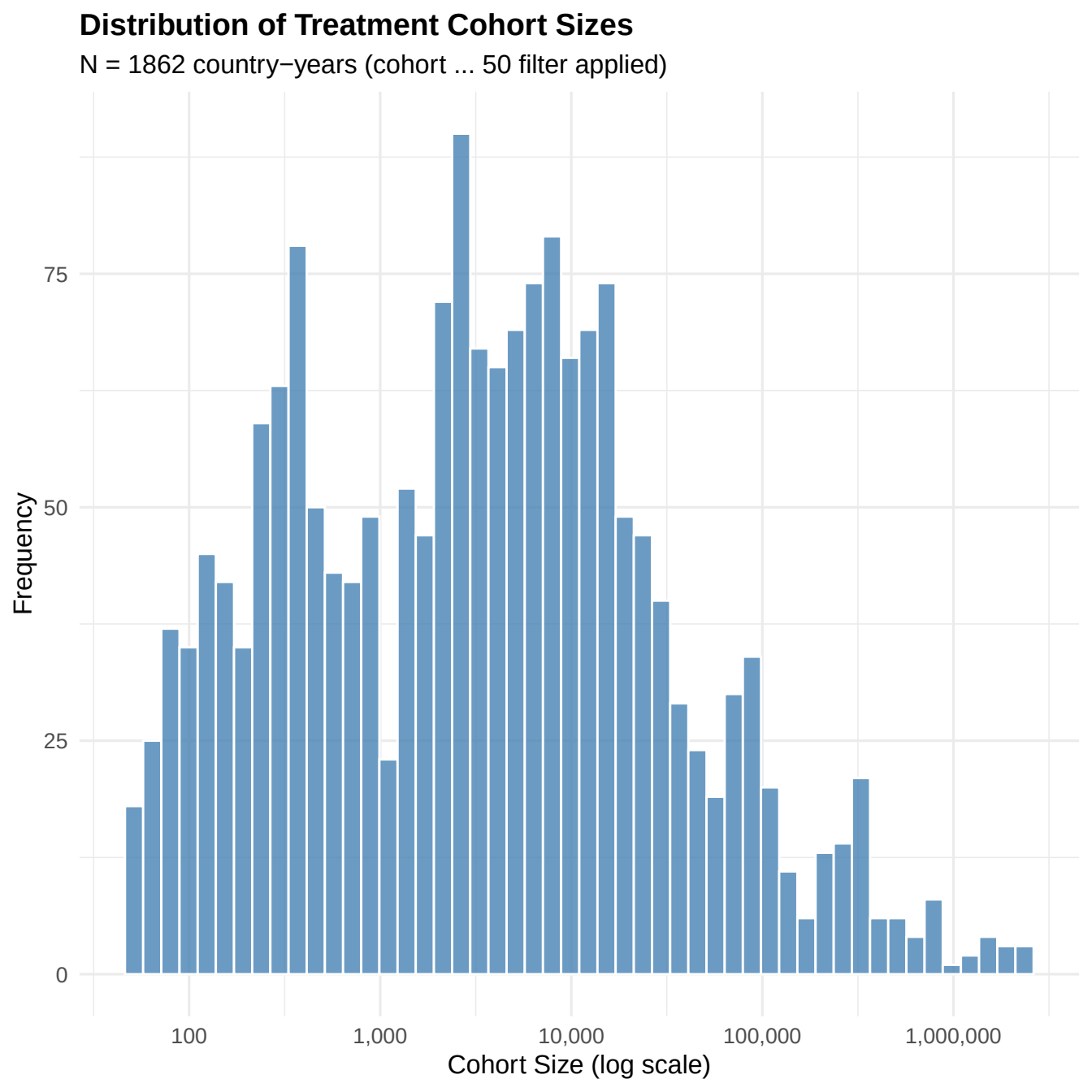

Outcome Data Availability by Year (2012...2023)
Before and after applying main inclusion flag (rel_with_new_flg = 1)
Number of country−year rows
200
180
160
140
2012
2013
2014
2015
2016
2017
2018
2019
2020
2021
2022
2023
Year
Condition
All rows
Has outcome (newrel_succ & newrel_coh)
Has outcome + rel_with_new_flg=1

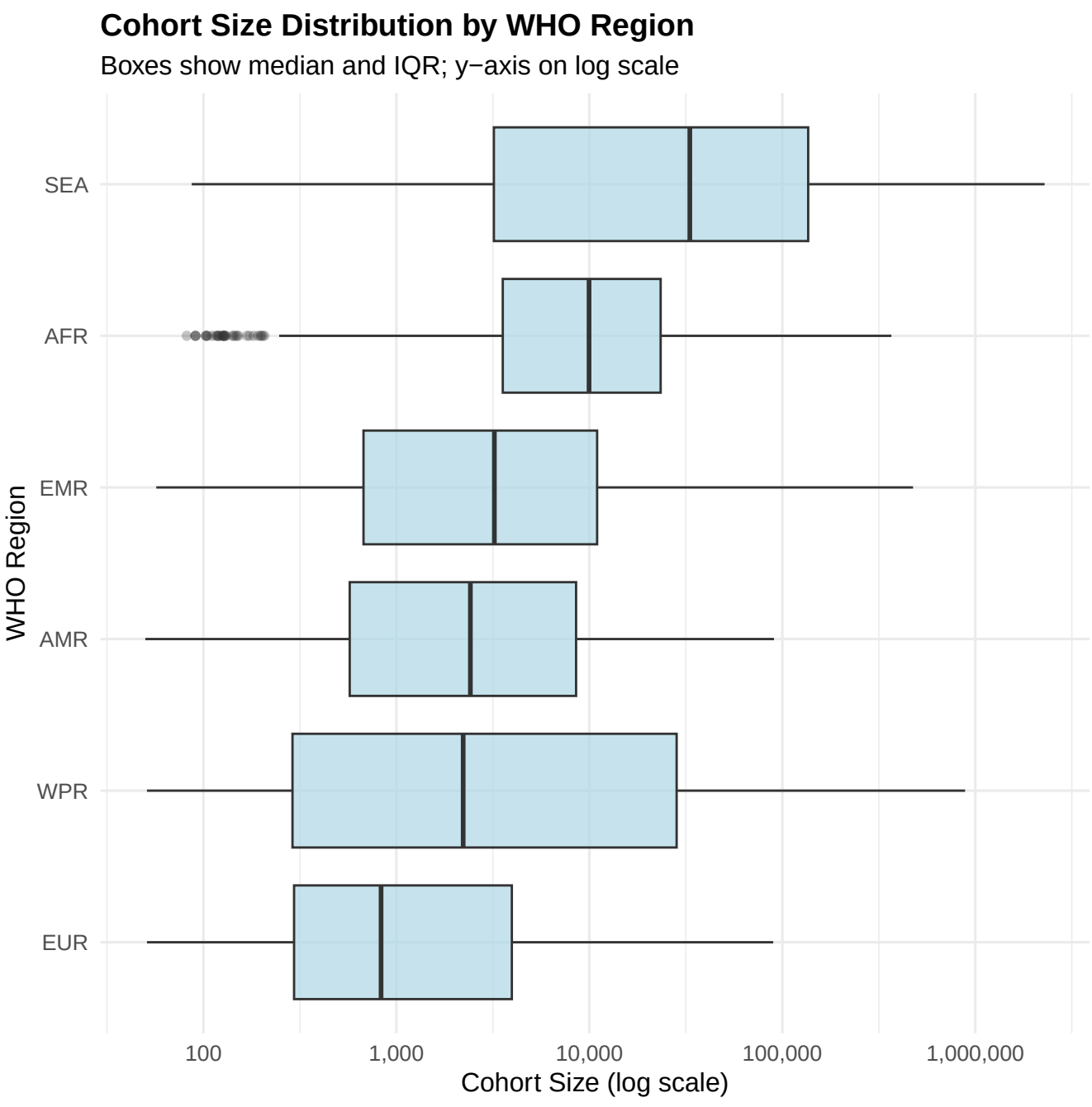
Data Completeness by Region and Year (2012−2023)

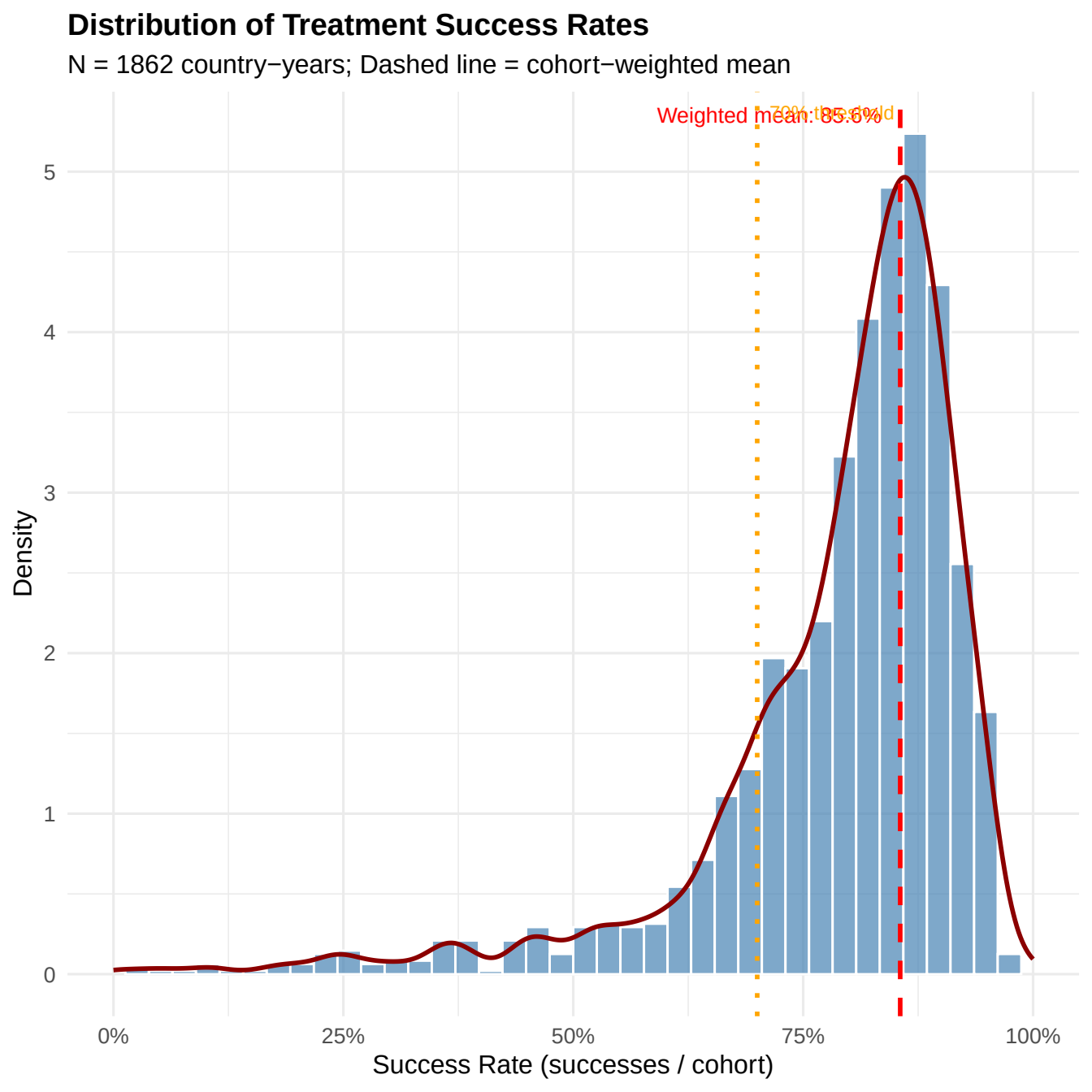
Complete = has outcomes, rel_with_new_flg=1, and core predictors

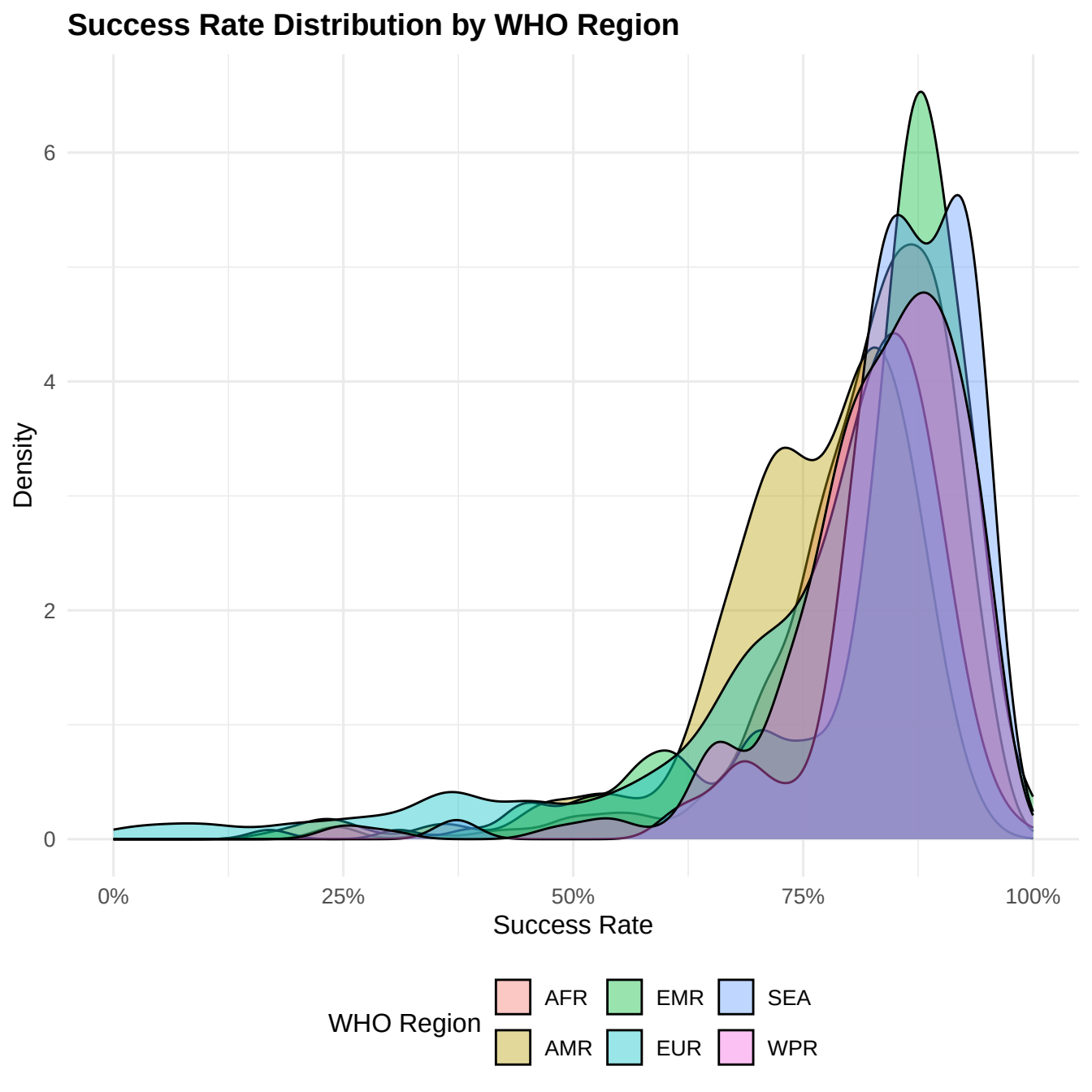


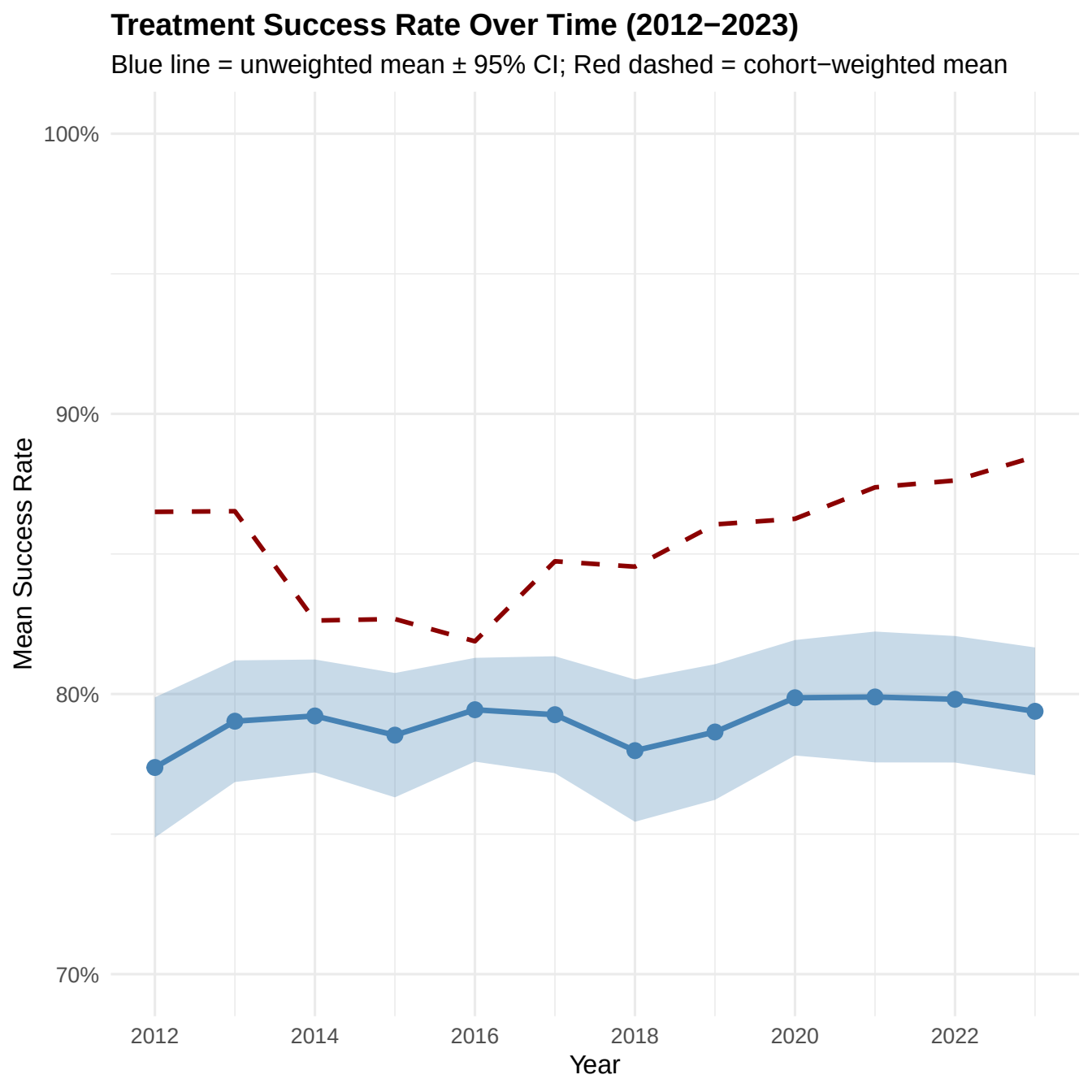

Impact of Cohort >= 50 Filter by WHO Region
Total: 209 rows lost (10.1%), 26 countries lost entirely
AMR
WPR
EUR
AFR
EMR
SEA
22.5%
20.7%
6.6%
2.5%
2.2%
0.0%
WHO Region
0
10
20
% Rows Lost

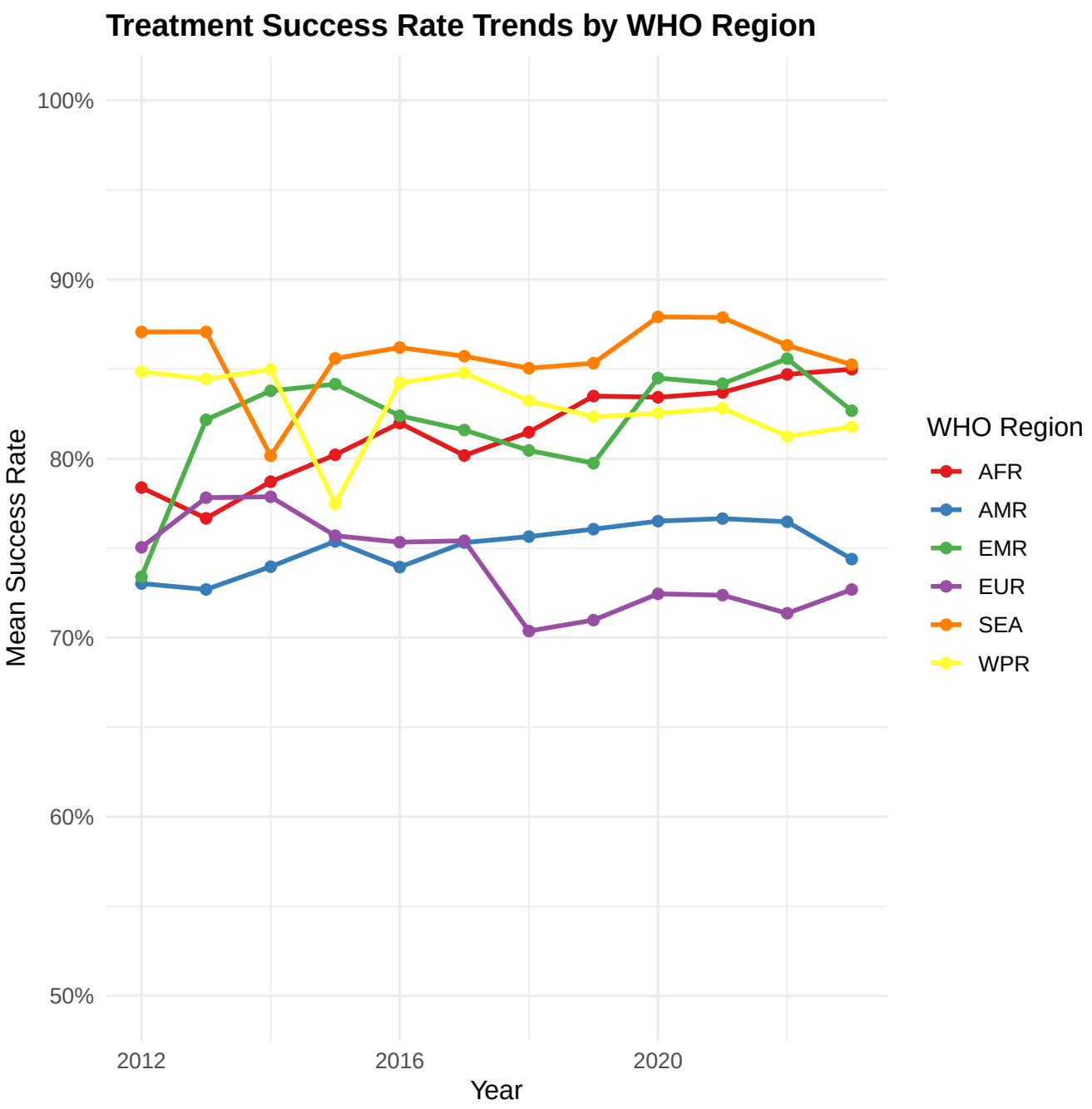

Distribution of Treatment Cohort Sizes
N = 1862 country−years (cohort ... 50 filter applied)
Frequency
0
25
50
75
100
1,000
10,000
100,000
1,000,000
Cohort Size (log scale)

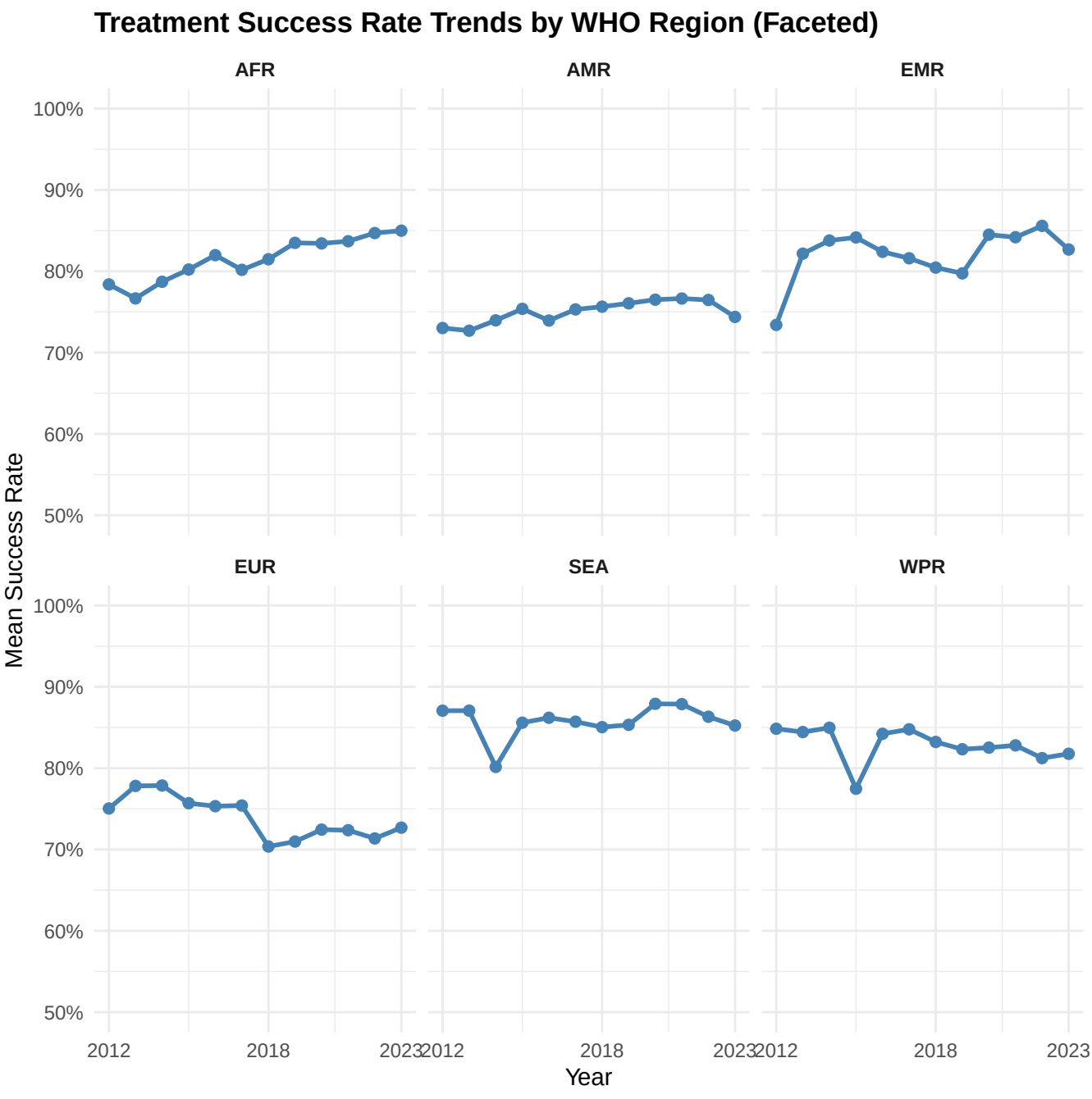

Cohort Size Distribution by WHO Region
Boxes show median and IQR; y−axis on log scale
SEA
AFR
EMR
AMR
WPR
EUR
WHO Region
100
1,000
10,000
100,000
1,000,000
Cohort Size (log scale)

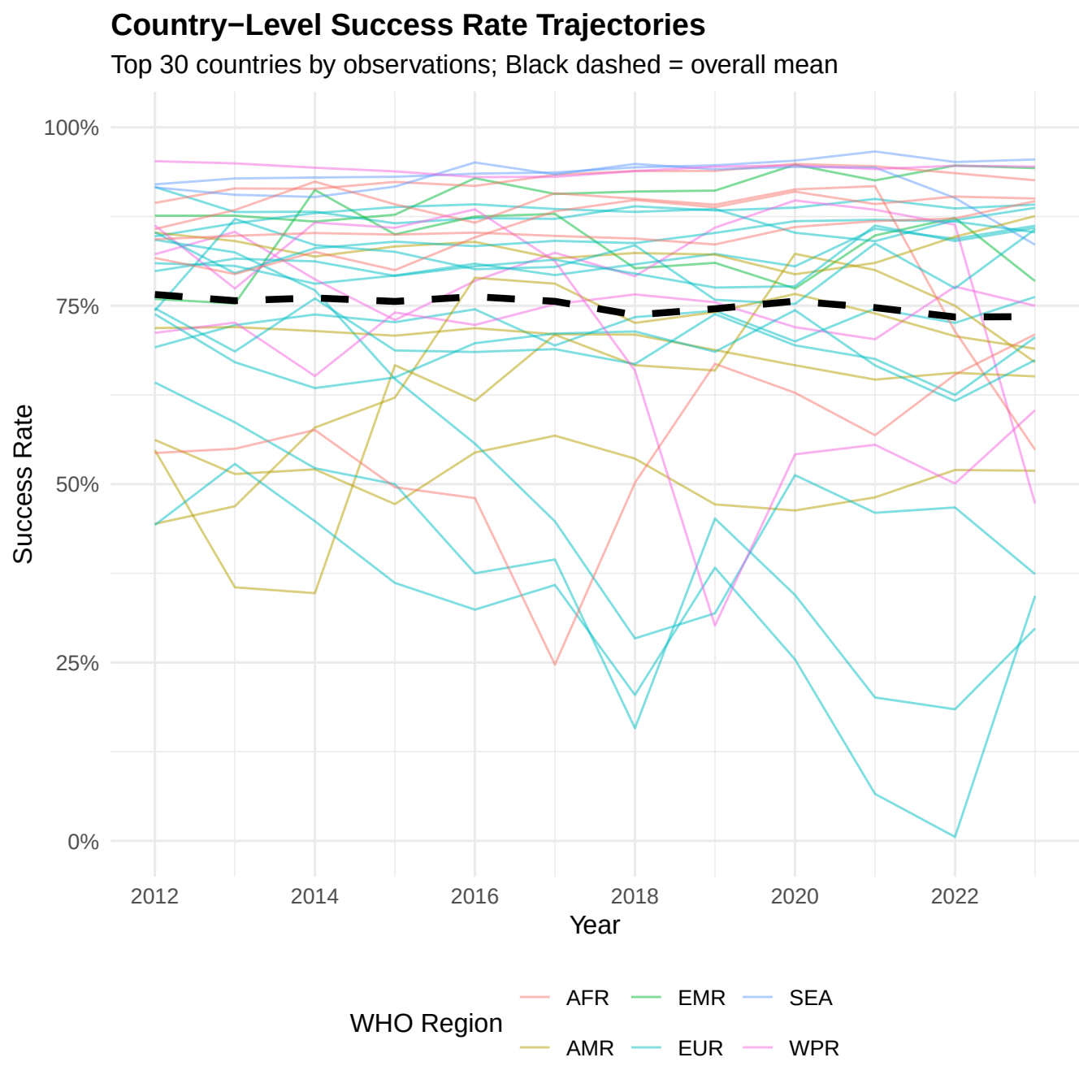

Distribution of Treatment Success Rates
N = 1862 country−years; Dashed line = cohort−weighted mean
Weighted mean: 85.6%
70% threshold
Density
0
1
2
3
4
5
0%
25%
50%
75%
100%
Success Rate (successes / cohort)


Success Rate Distribution by WHO Region
Density
6
4
2
0
0%
25%
50%
75%
100%
Success Rate
WHO Region
AFR
EMR
SEA
AMR
EUR
WPR

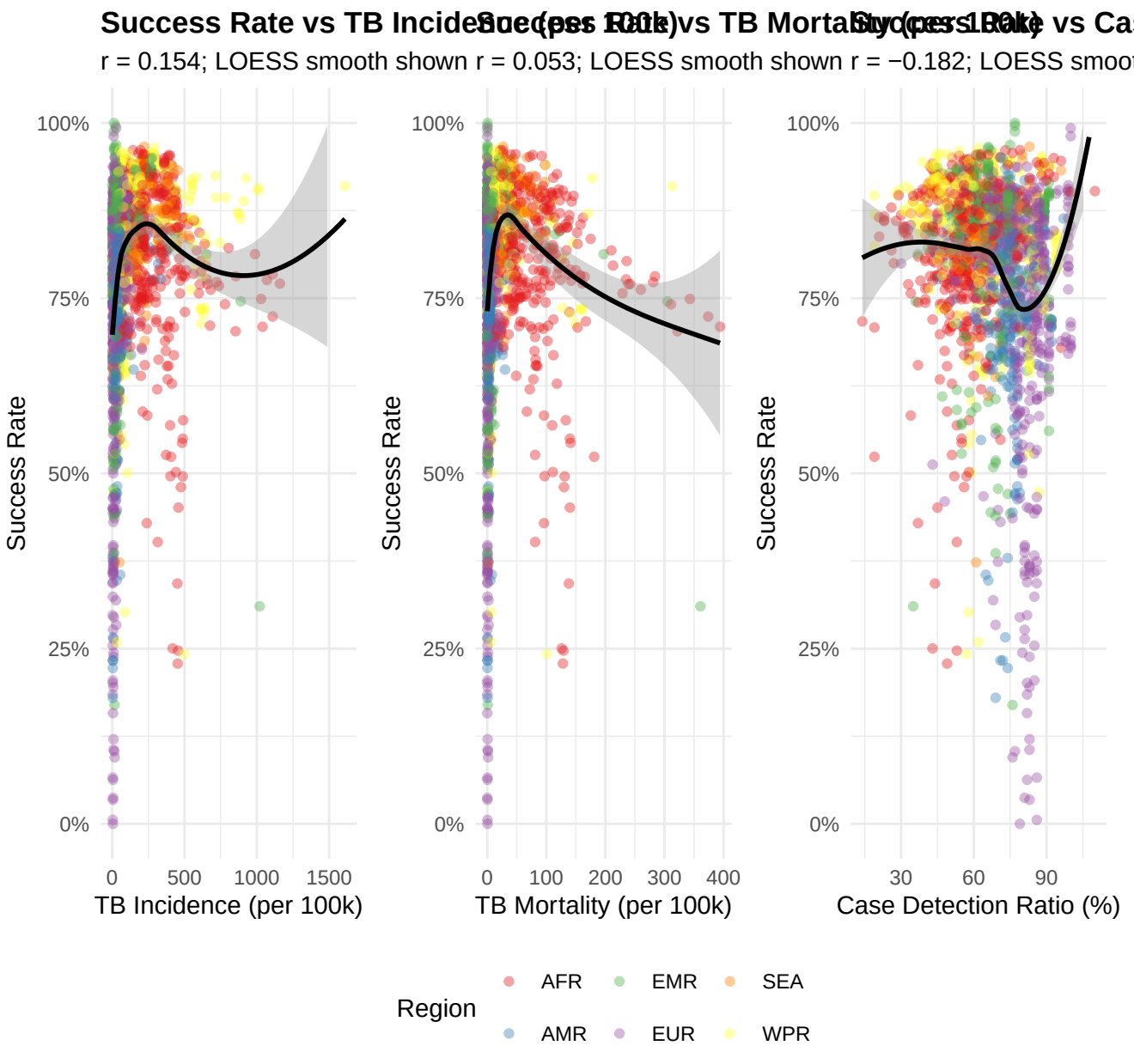

Treatment Success Rate Over Time (2012−2023)
Blue line = unweighted mean ± 95% CI; Red dashed = cohort−weighted mean
100%
90%
80%
70%
Mean Success Rate
2012
2014
2016
2018
2020
2022
Year

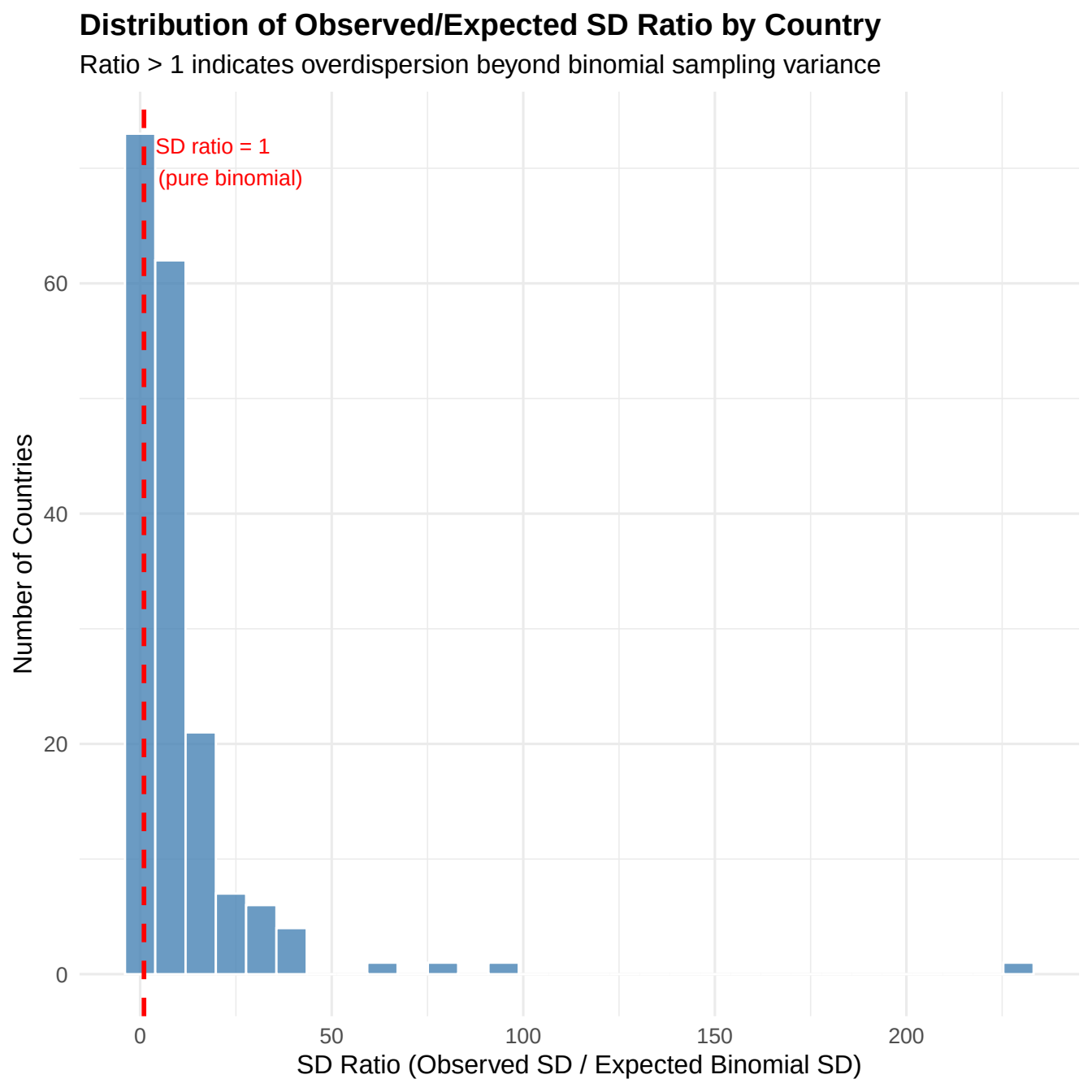

Treatment Success Rate Trends by WHO Region
100%
90%
80%
70%
60%
50%
Mean Success Rate
2012
2016
2020
Year
WHO Region
AFR
AMR
EMR
EUR
SEA
WPR

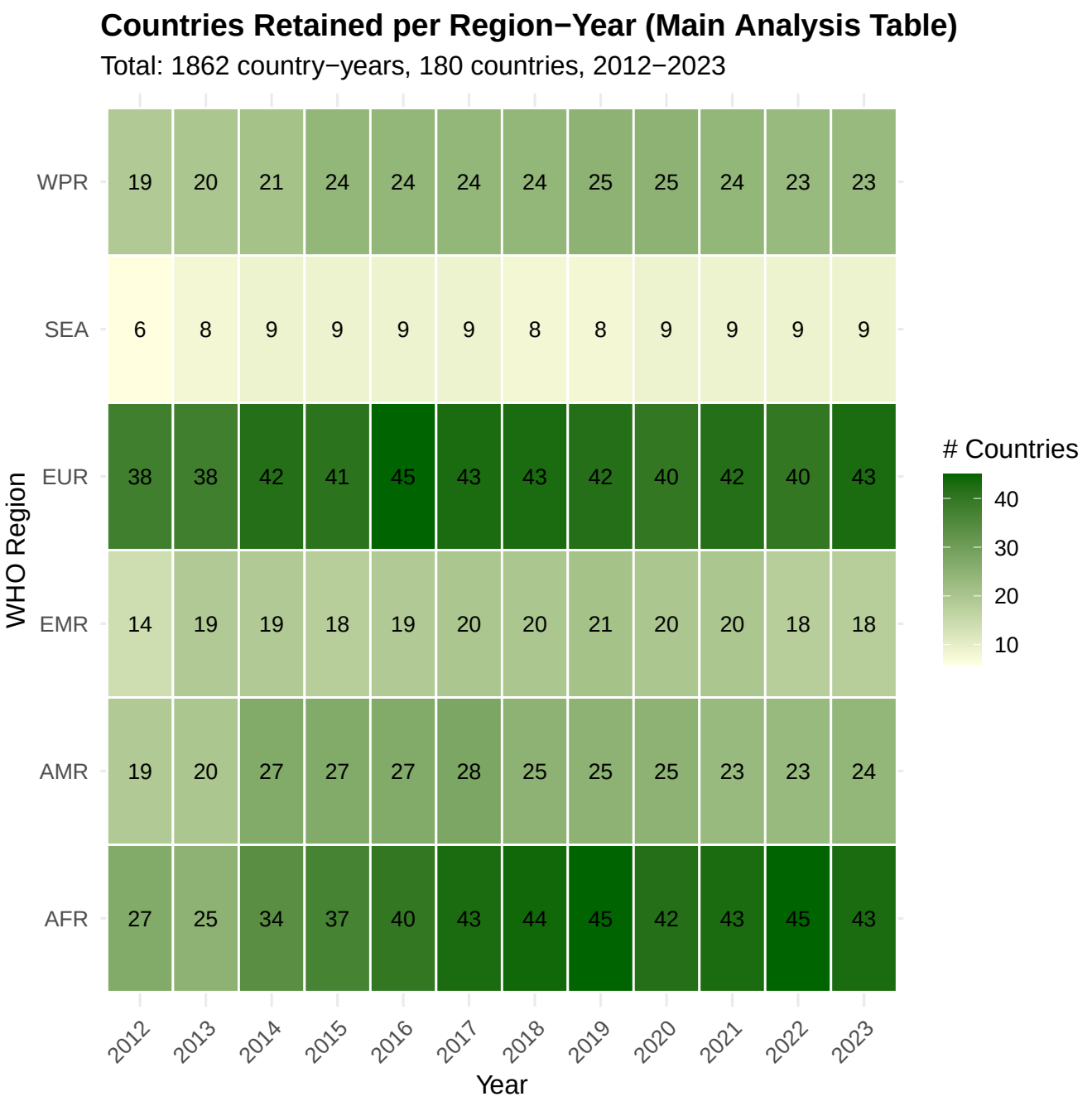

Treatment Success Rate Trends by WHO Region (Faceted)
AFR
AMR
EMR
EUR
SEA
WPR
Mean Success Rate
100%
90%
80%
70%
60%
50%
2012
2018
2023
Year

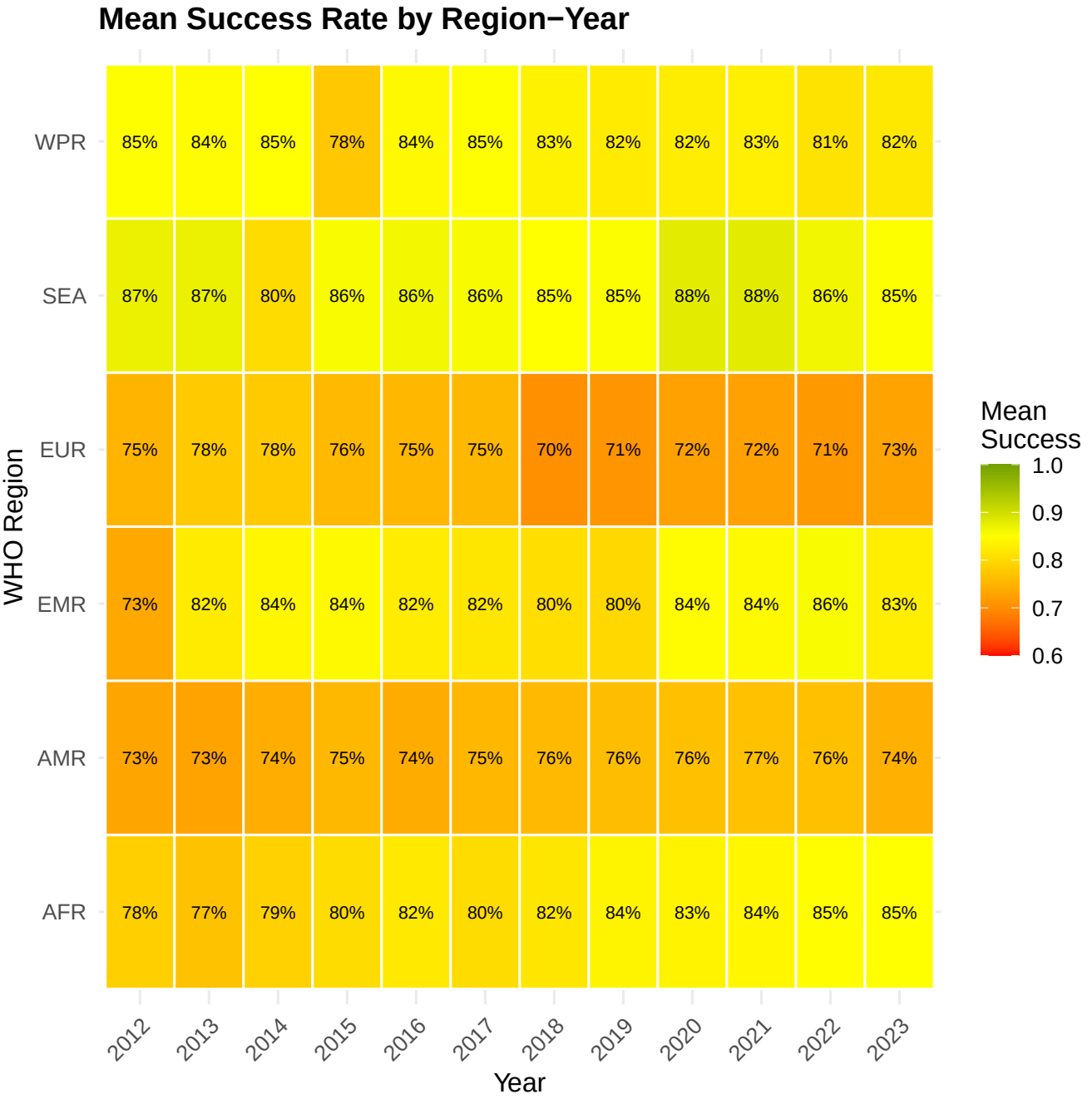

Country−Level Success Rate Trajectories
Top 30 countries by observations; Black dashed = overall mean
100%
75%
50%
25%
0%
Success Rate
2012
2014
2016
2018
2020
2022
Year
WHO Region
AFR
EMR
SEA
AMR
EUR
WPR

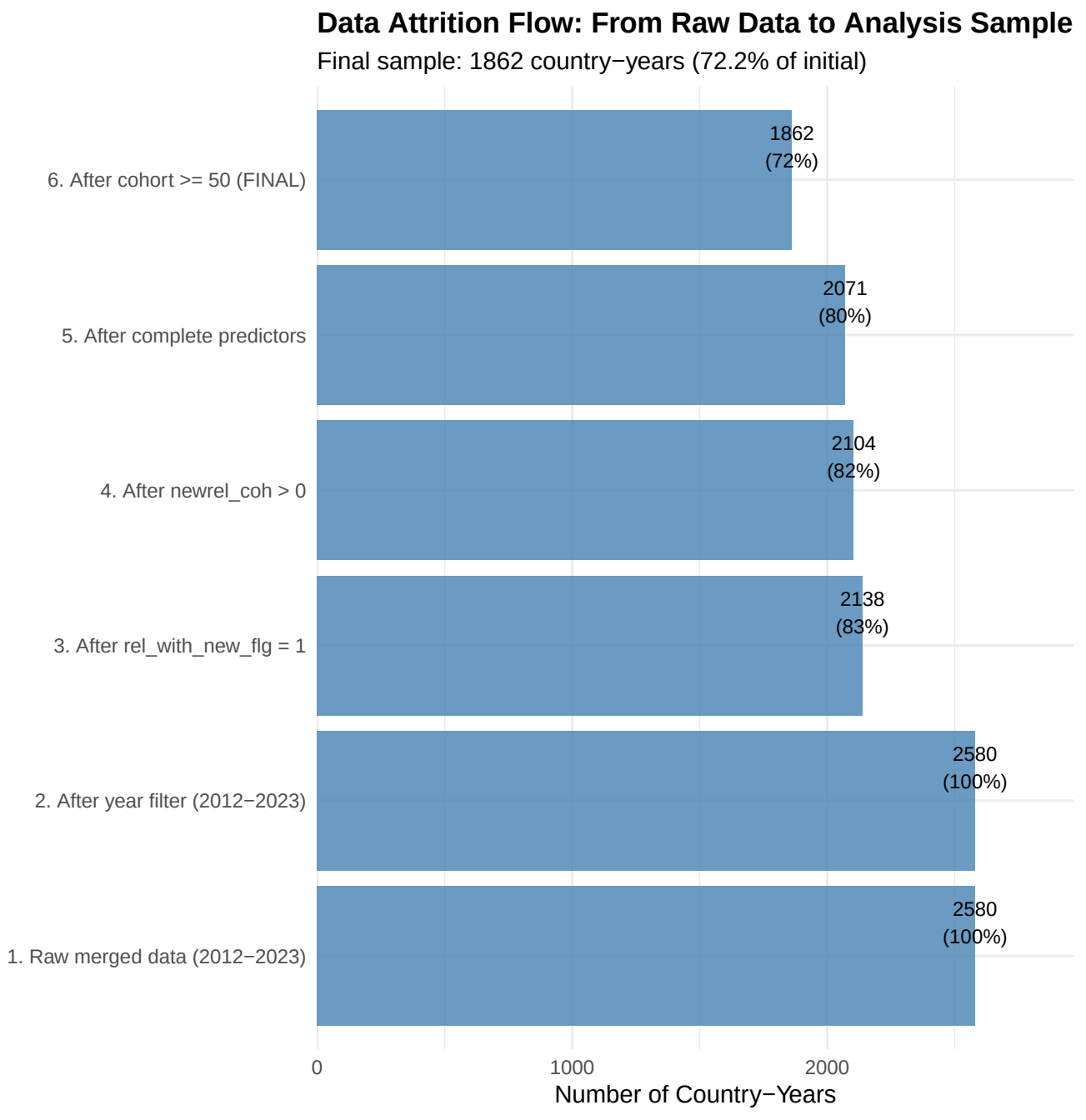
Bivariate Relationships: Predictors vs Treatment Success

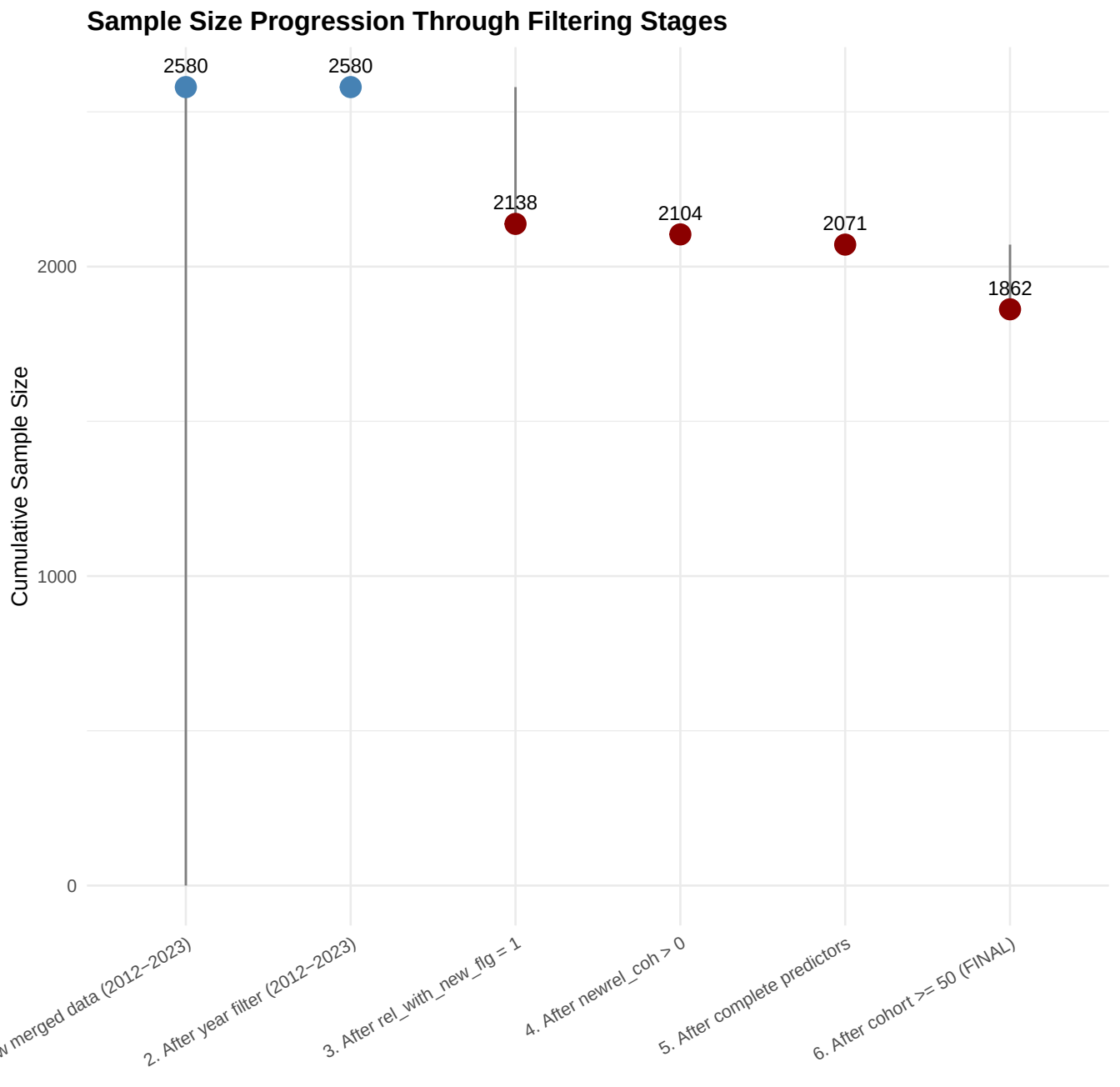


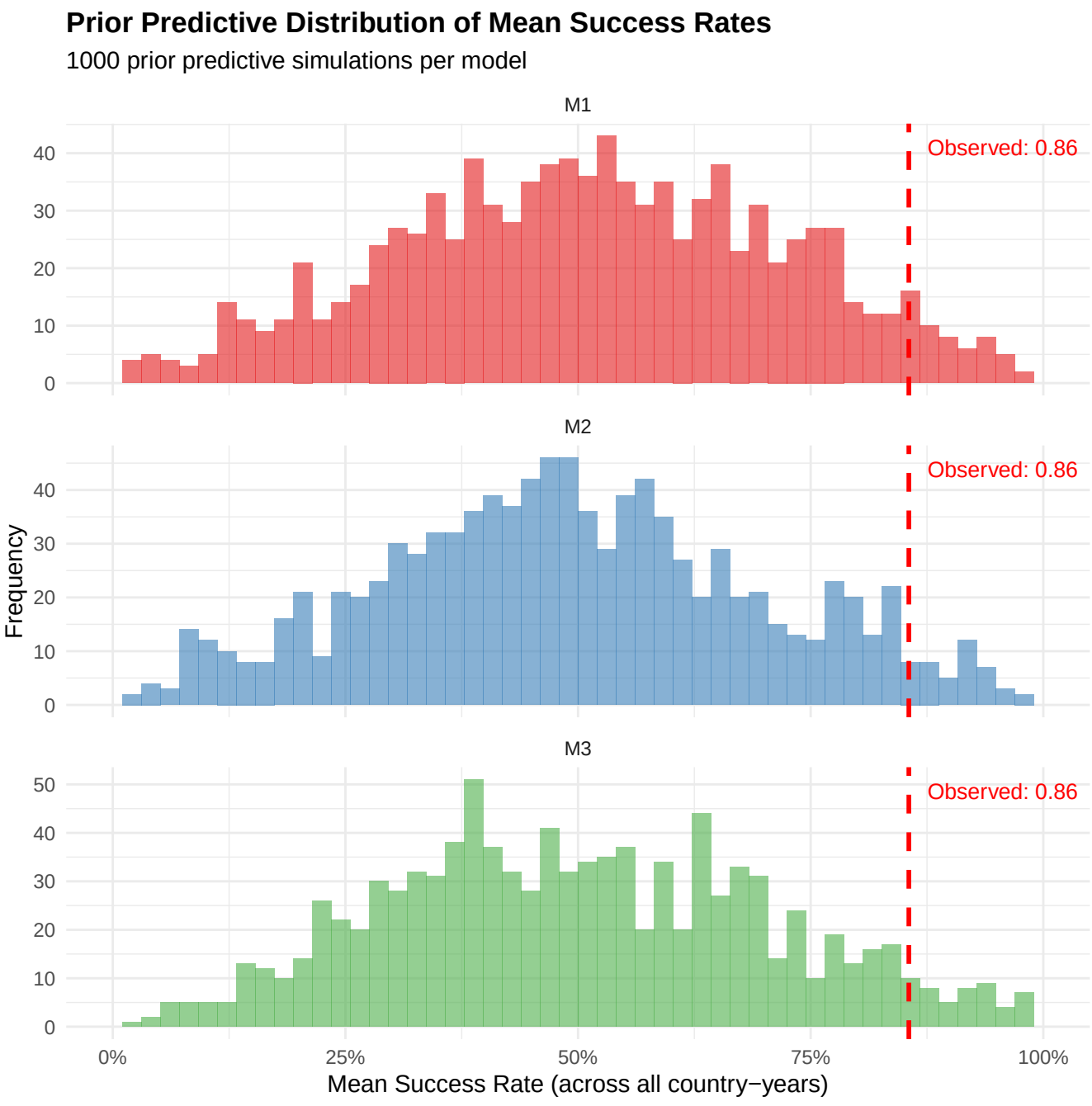

Distribution of Observed/Expected SD Ratio by Country
Ratio > 1 indicates overdispersion beyond binomial sampling variance
SD ratio = 1
(pure binomial)
Number of Countries
0
20
40
60
0
50
100
150
200
SD Ratio (Observed SD / Expected Binomial SD)


Countries Retained per Region−Year (Main Analysis Table)
Total: 1862 country−years, 180 countries, 2012−2023
WHO Region
WPR 19 20 21 24 24 24 24 25 25 24 23 23
SEA 6 8 9 9 9 9 8 8 9 9 9 9
EUR 38 38 42 41 45 43 43 42 40 42 40 43
EMR 14 19 19 18 19 20 20 21 20 20 18 18
AMR 19 20 27 27 27 28 25 25 25 23 23 24
AFR 27 25 34 37 40 43 44 45 42 43 45 43
2012 2013 2014 2015 2016 2017 2018 2019 2020 2021 2022 2023
Year
Countries
40
30
20
10

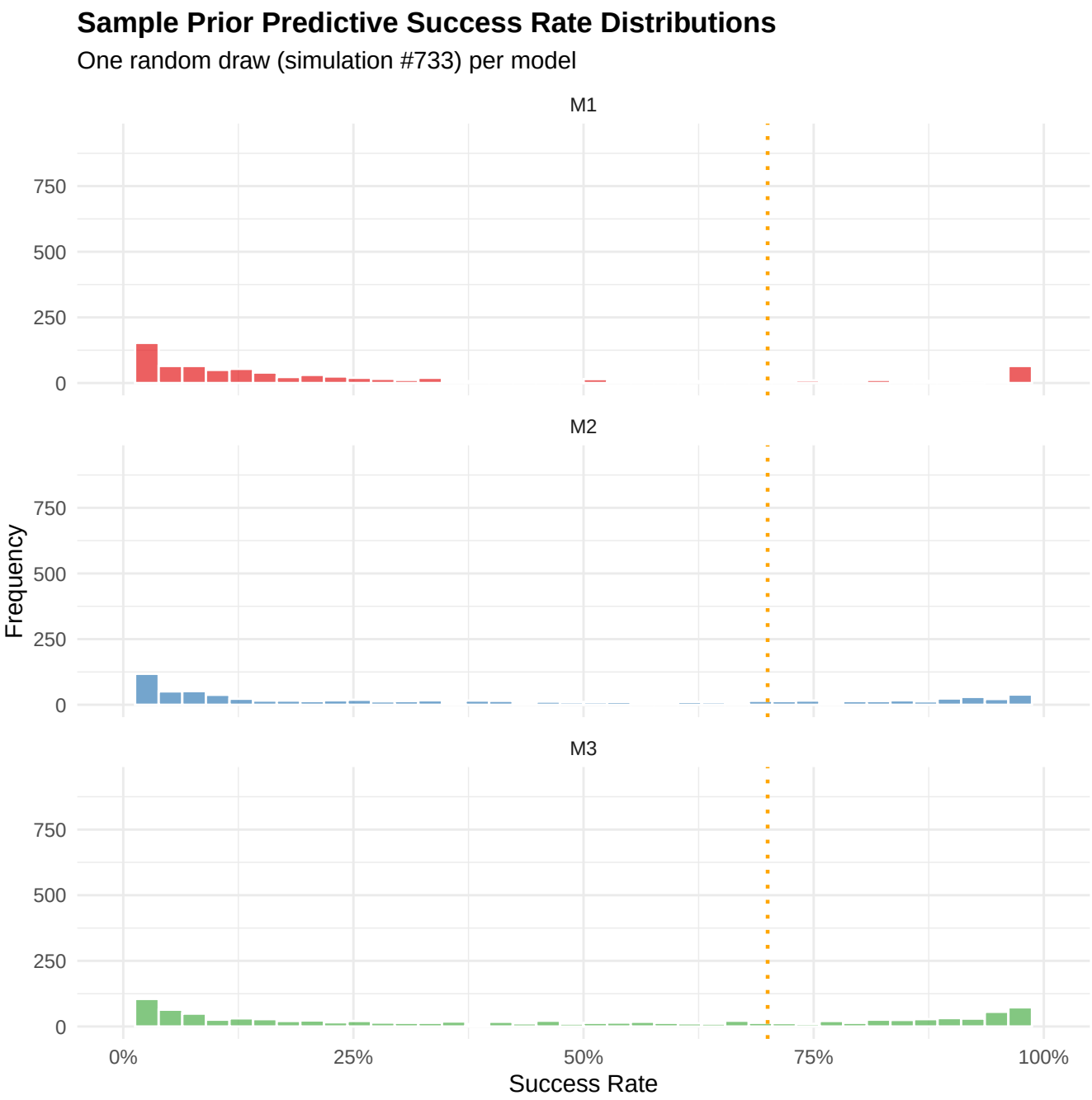

Mean Success Rate by Region−Year
WHO Region
WPR
SEA
EUR
EMR
AMR
AFR
85% 84% 85% 78% 84% 85% 83% 82% 82% 83% 81% 82%
87% 87% 80% 86% 86% 86% 85% 85% 88% 88% 86% 85%
75% 78% 78% 76% 75% 75% 70% 71% 72% 72% 71% 73%
73% 82% 84% 84% 82% 82% 80% 80% 84% 84% 86% 83%
73% 73% 74% 75% 74% 75% 76% 76% 76% 77% 76% 74%
78% 77% 79% 80% 82% 80% 82% 84% 83% 84% 85% 85%
2012 2013 2014 2015 2016 2017 2018 2019 2020 2021 2022 2023
Year
Mean Success
1.0
0.9
0.8
0.7
0.6


Data Attrition Flow: From Raw Data to Analysis Sample
Final sample: 1862 country−years (72.2% of initial)
6. After cohort >= 50 (FINAL)
1862
(72%)
5. After complete predictors
2071
(80%)
4. After newrel_coh > 0
2104
(82%)
3. After rel_with_new_flg = 1
2138
(83%)
2. After year filter (2012−2023)
2580
(100%)
1. Raw merged data (2012−2023)
2580
(100%)
0
1000
2000
Number of Country−Years


Sample Size Progression Through Filtering Stages
Cumulative Sample Size
2000
1000
0
2580
2580
2138
2104
2071
1862
w merged data (2012−2023)
2. After year filter (2012−2023)
3. After rel_with_new_flg = 1
4. After newrel_coh > 0
5. After complete predictors
6. After cohort >= 50 (FINAL)


Prior Predictive Distribution of Mean Success Rates
1000 prior predictive simulations per model
M1
Observed: 0.86
M2
Observed: 0.86
M3
Observed: 0.86
Frequency
0
10
20
30
40
50
0%
25%
50%
75%
100%
Mean Success Rate (across all country−years)

Prior Predictive Distribution of Success Rate SD
1000 prior predictive simulations per model
M1
Observed: 0.14
M2
Observed: 0.14
M3
Observed: 0.14
Frequency
0
25
50
75
0.0
0.1
0.2
0.3
0.4
0.5
SD of Success Rates


Sample Prior Predictive Success Rate Distributions
One random draw (simulation #733) per model
M1
M2
M3
Frequency
750
500
250
0
0%
25%
50%
75%
100%
Success Rate